

Vinaykumar Venkateshkumar

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UK Student visa, coursework complete · Available immediately · Sponsorship may be required

Profile

Graduate design engineer for the engine externals and secondary-systems lane: release-ready mechanical design from parametric CAD — detail and GA drawings to ISO 1101 and ISO 286, tolerance stack-up, flat patterns, DFM — sized by FEA. Writes the tolerance a result must meet *before* running it, and publishes the runs that miss it.

Technical Skills

Mech. Design	Geometric tolerancing (ISO 1101), limits and fits (ISO 286), tolerance stack-up (worst case and RSS), design for manufacture, sheet-metal forming (K-factor, bend allowance, flat patterns), detail and GA drawings
CAD	CATIA V5 (proficient, 80 hrs certified), CadQuery, pyOCC / Open CASCADE, ANSYS DesignModeler, STEP and DXF interchange, parametric geometry
Analysis	FEA (CalculiX, gmsh, FreeCAD FEM): load cases, mesh convergence, stress-singularity identification, hand calculation as an independent check. CFD (ANSYS Fluent, ICEM, OpenFOAM): $k-\omega$ SST, grid independence. Python (NumPy, SciPy, pytest), MATLAB, Git

Selected Projects

Parametric Actuator and Accessory-Gearbox Generators, with Drawing Pack — *mechanical design, GD&T* 2026

- Sized before drawn — a flight-control actuator family across four aircraft classes and an accessory gearbox across five power ratings (5–50 kW), with involute gear geometry, Lewis bending-stress sizing and SKF 60xx bearings — then issued as a four-sheet A4 pack: three details and a ballooned GA with ISO 286 fits (35 H8 bore, 21 f7 rod), ISO 1101 tolerances, a datum scheme and a five-contributor stack on 390 mm nominal (worst case ± 0.90 , RSS ± 0.44).
- Verification found six real defects, including meshing teeth intersecting by 424 mm^3 in every exported assembly because the gears were never phased — then the test suite was checked against itself with a reintroduced sign error, which only the tooth-taper test catches. 132 automated tests.

Formed Sheet-Metal Equipment Bracket, Sized and Redesigned — *sheet metal, DFM, FEA* 2026

- Flat pattern from a K-factor neutral axis rather than assumed (101.52 mm blank against 105.00 mm of summed legs), validated by conservation of volume to 0.25% against 5.12% for a naive blank; five DFM rules, and a material trade with a real answer — 2024-T3 is twice as strong as 5052-H32 and cannot make the fold, 4T bend radius against 1T.
- Sized against a 9g crash load in CalculiX, where a five-density convergence study made the result usable: peak stress climbed 12% and never settled — a singularity at a fixed constraint, so the 545 MPa headline is a boundary condition, not a stress, and the converged 196.7 MPa against 193 MPa yield condemns the part marginally rather than by $2.8\times$. Four fixes were screened by hand first — stiffening the upright makes that leg $7.4\times$ stiffer and changes the margin by nothing — before 2.0 mm gauge (126.1 MPa, +53%) and a re-issued drawing, the bend deduction having moved with the gauge. 66 tests.

Nacelle Cowl Geometry and CFD Validation — *personal project, public data* 2026

CST curve fitted to NASA TM 110300's published NACA 1-85-100 cowl ordinates to 0.019% RMS of max radius; a compressible RANS OpenFOAM case ($k-\omega$ SST, 108,904 cells) at Mach 0.79 matched to that report's wind-tunnel pressures to RMS C_p 0.65 over 17 stations, including where it misses the suction peak. *Separately: a Cranfield research project on nacelle installation aerodynamics — industrial, confidential, result pending.*

NASA/GE E³ Turbofan, Rebuilt from its Design Reports — *gas turbine performance, turbomachinery* 2026

- Nine of ten stages of a twin-spool turbofan rebuilt from public-domain NASA reports — cycle, blading, cooling, secondary air, blade and disc stress, mass — every value carrying its report page and every solver stating the tolerance it must meet *before* it runs: HPC efficiency lands 0.15 of a point from the published value on a 2.0-point band, while engine sfc lands at 1.91% against a 1.5% band and is recorded as a miss, not rounded into a pass. Over 1,000 automated tests. Published as an interactive 3D anatomy at vinaykumar.is-a.dev/turbofan — 144 procedural parts across 12 systems, blade rows lofted from printed section tables rather than drawn by eye, every fact tagged to a report page or flagged schematic or assumed, with a six-sheet drawing pack behind it.
- ME TF0.01** — a derivative trade *study* of my own on that fixed core, never built: what bypass ratio 10 costs. Fan and LP turbine want the LP shaft 46% apart at BPR 10, against 0.7% at the E³'s own 6.7 — the geared-fan argument as a number, and on this core the answer is more LP turbine stages, not a 1.46:1 gearbox. The tip-Mach ceiling written down before the run rejected my own first answer of seven stages; it needs eight. Fuel burn is reported as a 3–4.5% band rather than the 4.5% headline, because fan efficiency is held at the E³'s values at a different tip Mach — worth 0.68% of sfc per point.

Education

Postgraduate Coursework, Thermal Power and Propulsion — Cranfield University, Cranfield, UK Oct 2025 – Sep 2026

Gas Turbine Performance · Combustion · Turbomachinery Aerodynamics · Propulsion System Design

BEng Aeronautical Engineering — CGPA 7.37/10 (first class) — KCG College of Technology, Chennai, India 2021 – 2025

Certifications: CATIA V5 (Proficient), CADD Centre, 80 hrs, 2024 · IELTS Academic 6.5 (UKVI B2), 2025

Experience

Engineering Intern, Aircraft Maintenance — AIESL — Boeing 737 MRO base, India Mar – Apr 2025

Rotation across Component Overhaul, Production Planning and Stores servicing Air India Express. Documented wheel/brake overhaul to specification (158 lb-ft torque, 205 ± 5 psi) and Eddy Current Testing; tracked parts in AMOS and RAMCO.